
INTERNSHIP: Tcs-ion-Rio-125--Forecasting-Retail-Demand

Internship Project Title	Forecasting System - Project Demand of Products at a Retail Outlet Based on Historical Data
Name of the Company	TCS iON
Name of the Industry Mentor	Harish Kumar
Name of the Institute	ICT Academy of Kerala

Start Date	End Date	Total Effort (hrs.)	Project Environment	Tools used
04.02.2023	.03.2023	95 hours (28 day)	Jupyter notebook	Python 3 (Numpy, Pandas, Matplotlib, Seaborn)

Acknowledgement

I would like to express my deepest gratitude to Mr. Harish Kumar and TCS ion for providing me with the necessary facilities for this project. I would like to express my sincere thanks to ICT Academy for making me capable for the internship by providing basic lessons. I would like to express my sincere thanks to the active discussion room in this project and valuable suggestions.

Objective

The objective is to build a forecasting system to predict the demand of a products at a retail outlet based on historical data by creating a forecast model, applying concepts of moving averages, forecasting methods, ARIMA models and time series forecasting. The model should be able to predict future sales by training on the historical sales data.

Introduction

Forecasting is a technique of predicting the future based on the results of previous data. Realization of the fact that "Time is Money" in business activities, the dynamic decision technologies presented here, have been a necessary tool for applying to a wide range of managerial decisions successfully where time and money are directly related. In making strategic decisions under uncertainty, we all make forecasts. It involves a detailed analysis of past and present trends or events to predict future events. The selection and implementation of the proper forecast methodology has always been an important planning and control issue for most firms. Forecasting is an important problem that spans many fields including business and industry, government, economics, environmental sciences, medicine, social science, politics, and finance.

The data set consists of sale's dates, product items, different stores and sales in a time span from 2013 to 2017. The task of this paper is to accurately predict the sales data of 2018 based on the previous data. The characteristic of this case is that different products are distributed in different stores, and at the same time, the sales data has obvious seasonality. The data has significant discrete features.

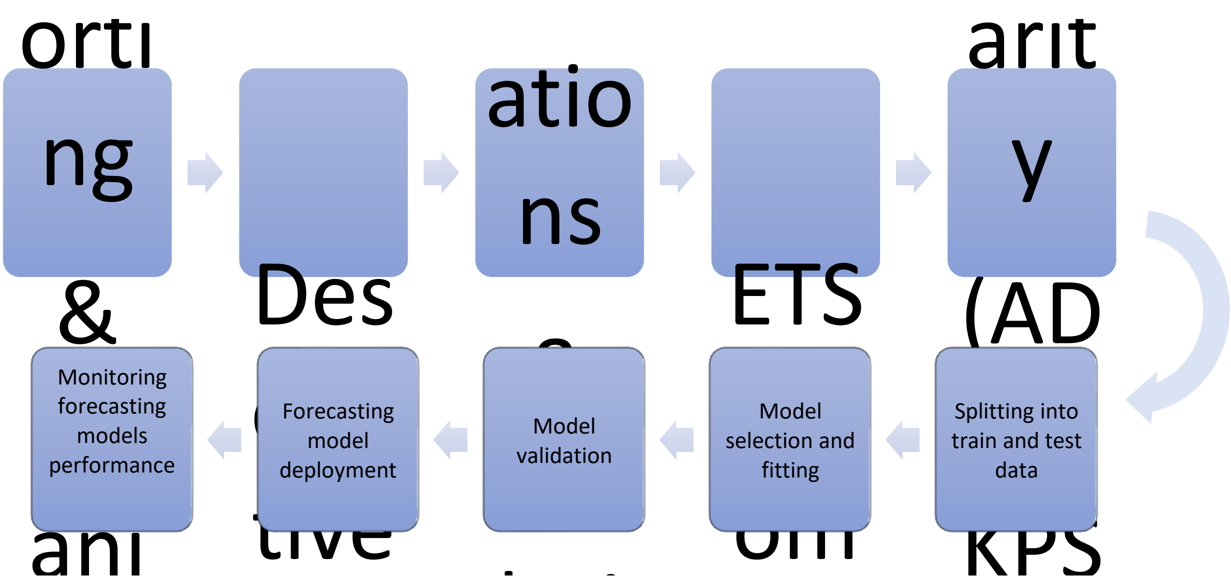
Internship Activities

- ✓ After creating an account in TCS Digital Learning Hub gone through all the contents in welcome kit and day wise plan.
- ✓ Attempted RIO pre-assessment and passed it successfully.
- ✓ Gone through you tube videos on ‘Forecasting Methods Overview’, ‘Moving Averages’, ‘Time series forecasting’, and ‘ARIMA models’ given in the project reference material.
- ✓ Watched webinar 1 and 2.
- ✓ Downloaded the data set from the given link in digital discussion room.
- ✓ Started working on the data set with Jupyter Notebook.
- ✓ Imported the data set and the necessary libraries.
- ✓ Made sure that the data set doesn’t contain any missing values.
- ✓ Gone through some basic information of the data set and features like shape,data types,count etc.
- ✓ Checked some descriptive statistics of the features like count, mean, median, standard deviations, quantiles, percentiles, min, max etc by grouping the features.
- ✓ Converted the ‘date’ feature to ‘datetime’ and created new features of year,month and day from given dates which is object type earlier.
- ✓ Plotted the total sales data of each stores, years,months and items separately.
- ✓ Separately plotted the total sales data of each of the stores occurring yearly,quarterly,monthly and weakly.
- ✓ Plotted histograms of sales of each stores to know the distribution patterns.
- ✓ Plotted heatmaps of sales of each months in every years for ten stores.
- ✓ Plotted the mean sales data of each of the product categories occurring on each month.
- ✓ Created boxplots based on sales of each months.
- ✓ Performed ETS (Error Trend Seasonality) Decomposition on sales data of each stores.
- ✓ Conducted Augmented-Dickey-Fuller test to verify which all stores are stationary.

- ✓ Conducted KPSS test to verify which all stores are stationary.

- ✓ Checked auocorrelation of each stores by ACF and PACF and plotted correlograms.
- ✓ Built the SARIMA model by finding best order and seasonal order.
- ✓ Created time series models of this order with the help of SARIMAX function.
- ✓ Done predictions of the data points in the unknown future.
- ✓ Compared predicted results with the test set data points and evaluated the performance of my model with the help of root mean square function.
- ✓ Compared the sales happening in each stores before and after forecasting and plotted them.
- ✓ Done time series modeling with Facebook's Prophet library.
- ✓ Made relevant plots of prophet model forecasting including trend and seasonality.
- ✓ Evaluated the performance of prophet model with the help of root mean square function.
- ✓ Compared all rmse scores of each stores of prophet model.
- ✓ Reached to a conclusion.

Methodology



Time-series data refers to data where the magnitude of changes over time, meaning that each data point represents a specific point in time and the corresponding value of the observed phenomenon at that time. **THE FORECASTING PROCESS** is a series of connected activities that transform one or more

inputs into one or more outputs.

Generally activities in the forecasting process are:

1. Problem definition
2. Data collection
3. Data analysis
4. Model selection and fitting
5. Model validation
6. Forecasting model deployment
7. Monitoring forecasting model performance

Defining the problem and understanding the target variable that we want to forecast is the first process to fix the specific goals . For that we should collect and clean the data that will be used for the time series forecast. This may include historical data for the target variable, as well as other related variables that may influence the forecast.

Later analyze the data to identify any trends, patterns, or seasonality in the time series. This may involve data visualization, descriptive statistics, or time series decomposition techniques. Visualisation is the best way to understand time-series data; converting data points into graphs can give us the overall overview. Sometimes preparing the data for the forecasting model by scaling the data, transforming the data, or removing any outliers or missing values takes much important role.

Choose a forecasting method

There are several types of time series forecasting models, some of the most common ones are:

1. ARIMA (Auto regressive Integrated Moving Average):

ARIMA is a popular model for time series forecasting. It is a class of models that captures both the auto correlation and the trend in the data.It is a class of models that includes three types of parameters: auto regressive (AR) parameters, difference (I) parameters, and moving average (MA) parameters.

The AR parameters capture the linear relationship between the values of the time series and its past values, while the MA parameters capture the linear relationship between the values of the time series and the errors in the past observations. The I parameter represents the degree of difference needed to make the time series stationary. The ARIMA model is specified using three integers denoted as (p, d, q), where:

p is the order of the AR model, which represents the number of lagged values of the time series to include in the model.

d is the degree of difference required to make the time series stationary.

q is the order of the MA model, which represents the number of lagged errors to include in the model.

However, ARIMA models are limited in their ability to handle more complex patterns in time series data such as seasonality. In such cases, a seasonal ARIMA (SARIMA) model or other more advanced time series models may be more appropriate.

2. SARIMA (Seasonal Auto regressive Integrated Moving Average) is a popular model for time series forecasting that is an extension of the ARIMA model. It takes into account not only the auto correlation and trend in the data but also seasonal patterns.The SARIMA model is defined by three sets of parameters:

Auto regressive (AR) parameters: These parameters capture the linear relationship between the values of the time series and its past values.

Integrated (I) parameter: This parameter captures the degree of differencing needed to make the time series stationary.

Moving average (MA) parameters: These parameters capture the linear relationship between the values of the time series and the errors in the past observations.

In addition to these parameters, SARIMA models also have seasonal parameters. The seasonal pattern is usually defined by the seasonal period, which is the number of time steps between seasonal peaks. The seasonal pattern can be captured by adding additional auto regressive, difference, and moving average parameters to the model. These are denoted by the lowercase letters "p", "d", and "q" along with an uppercase "P", "D", and "Q" to denote the seasonal orders.SARIMA models can be used to forecast time series data with seasonality and trends, and they can be especially useful for data that exhibit complex patterns of seasonality. The choice of parameters depends on the specific characteristics of the data and can be determined using various techniques, such as grid search or information criteria.

3. Prophet is a popular time series forecasting algorithm developed by Facebook. It is designed to handle seasonality, trends, and other time series characteristics. The algorithm is based on a decomposable time series model with three main components: trend, seasonality, and holidays. Prophet is particularly useful for forecasting data with multiple seasonality patterns or with data that have many missing values. It also provides a straightforward method for incorporating external regressors such as weather data or marketing campaigns.

There are many other time series forecasting models as well, but these are some of the most commonly used ones. The choice of model depends on the characteristics of the data and the specific problem being addressed. Model selection and fitting consists of choosing one or more forecasting models and fitting the model to the data. Evaluate the performance of the forecasting model on the test data may involve metrics like mean absolute error, mean squared error, or R-squared. Then use the forecasting model to make predictions for the future time period. Continue to monitor the performance of the forecasting model over time and update the model as new data becomes available. Then we can communicate the results of the time series forecasting project to stakeholders in a clear and concise manner.

Understanding the Data

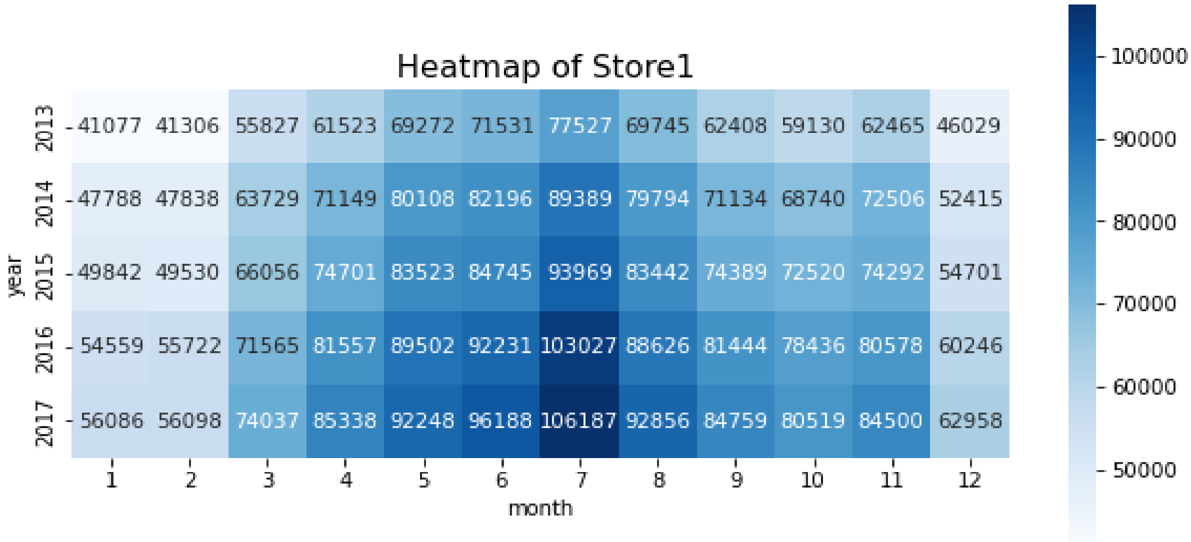
The first step is to load the data and transform it into a structure that we will then use for each of our models. In its raw form, each row of data represents a single day of sales at one of ten stores. Our goal is to predict monthly sales, so we will first consolidate all stores and days into total monthly sales. The first step is to load the data and transform it into a structure that we will then use for each of our models. In its raw form, each row of data represents a single day of sales at one of ten stores. Our goal is to predict monthly sales, so we will first consolidate all stores and days into total monthly sales.

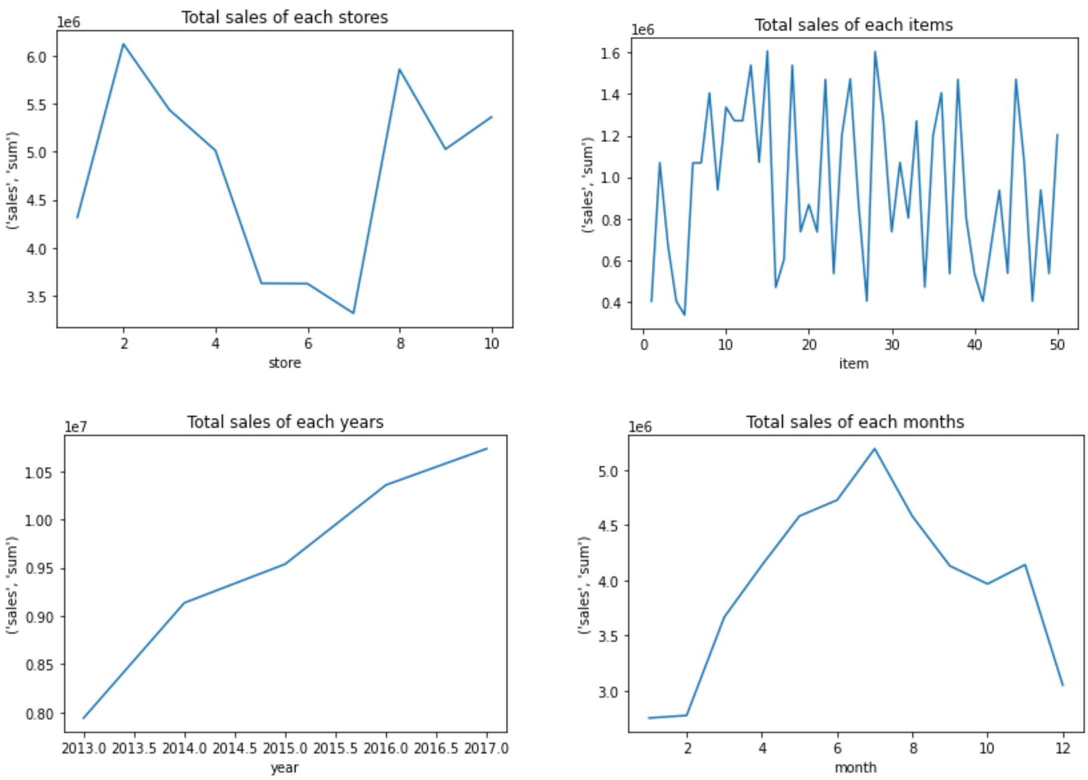
	date	store	item	sales
0	2013-01-01	1	1	13
1	2013-01-02	1	1	11
2	2013-01-03	1	1	14
3	2013-01-04	1	1	13
4	2013-01-05	1	1	10

Here we can see the data where we have got a column on date, stores, items and a sales column. The chain has 10 stores and 50 assortment items. The time span of the data is from 2013 to 2017 with values in date's column in string objects. We will be required to change it on the Date-Time object to create separate date features of year, month and day.

	date	store	item	sales	month	day	year
0	2013-01-01	1	1	13	1	1	2013
1	2013-01-02	1	1	11	1	2	2013
2	2013-01-03	1	1	14	1	3	2013
3	2013-01-04	1	1	13	1	4	2013
4	2013-01-05	1	1	10	1	5	2013

From the below heat map it is possible to quickly identify the months with highest or lowest sale in each year. The varying intensity of color represents the measure of sales in each months from 2013 to 2017. As in this example of store 1, we can identify the sales of other stores as well.

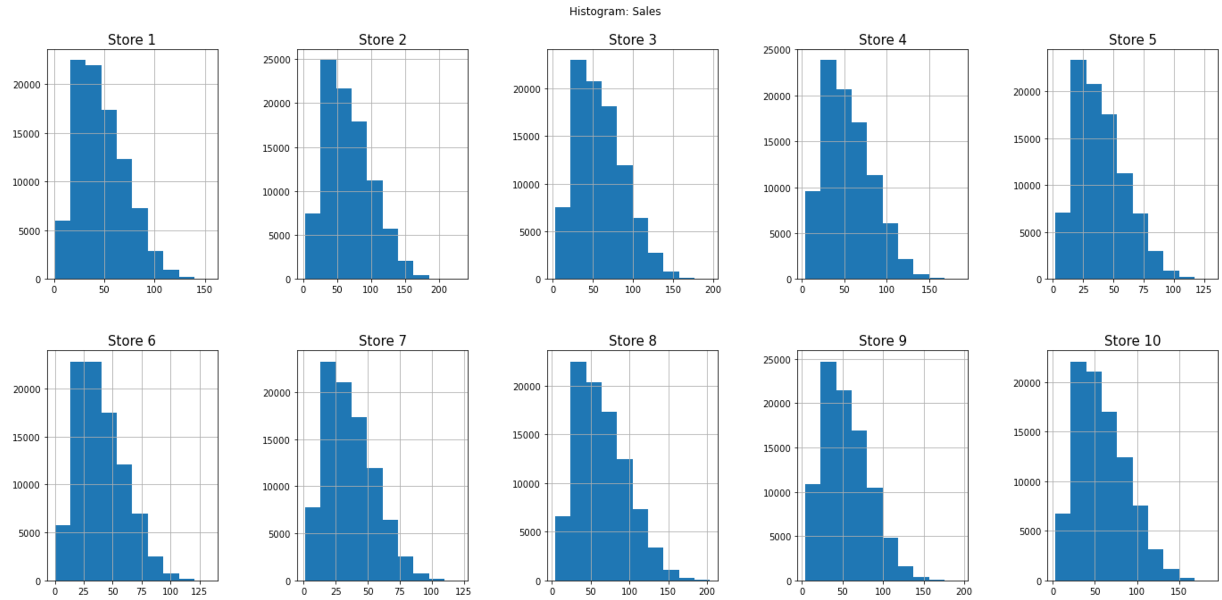

Heatmap of Store1
year
month
2013 41077 41306 55827 61523 69272 71531 77527 69745 62408 59130 62465 46029
2014 47788 47838 63729 71149 80108 82196 89389 79794 71134 68740 72506 52415
2015 49842 49530 66056 74701 83523 84745 93969 83442 74389 72520 74292 54701
2016 54559 55722 71565 81557 89502 92231 103027 88626 81444 78436 80578 60246
2017 56086 56098 74037 85338 92248 96188 106187 92856 84759 80519 84500 62958
1 2 3 4 5 6 7 8 9 10 11 12
100000
90000
80000
70000
60000
50000


Total sales of each stores
1e6
('sales', 'sum')
store
Total sales of each items
1e6
('sales', 'sum')
item
Total sales of each years
1e7
('sales', 'sum')
year
Total sales of each months
1e6
('sales', 'sum')
month

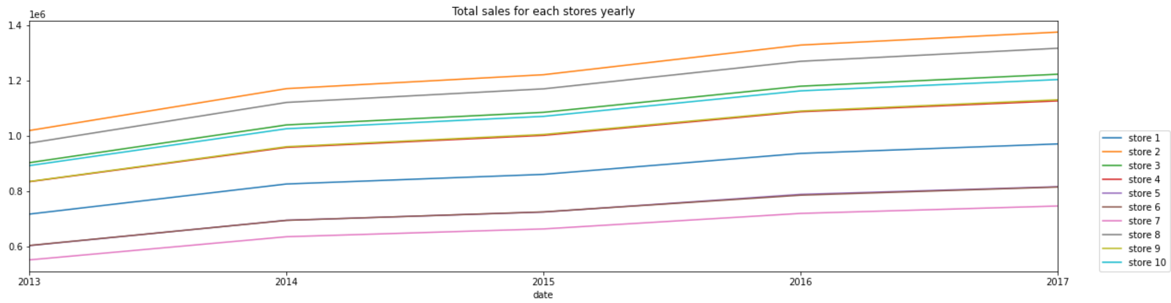
- From the store wise sales, the highest sales occurred in store 2 and the least sales occurred in store 7.
- There are some items with a high sale as item 15 or 29 and also there are some with very low sale as item 3 or 41.

- When analyzing the year wise trend of total sales, the increasing trend from 2013 till the end is very significant.
- In the above plot of total sales, it can be seen that there is an increasing trend in the periods of January to July. Then we can see a decreasing trend till December with a slight exception on November.

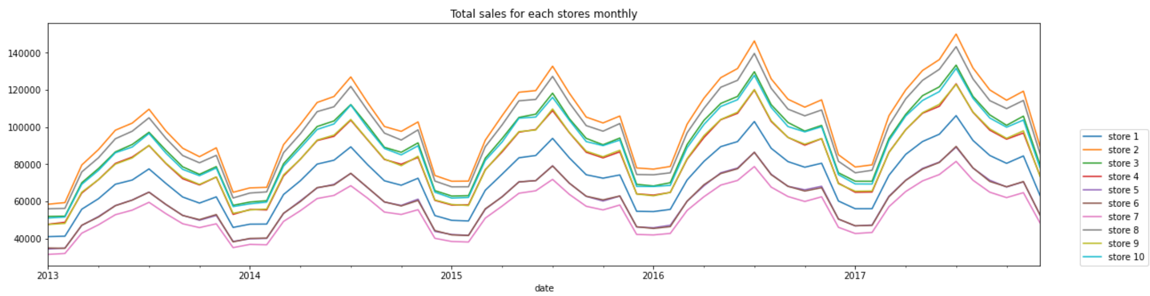
The histograms of sales:

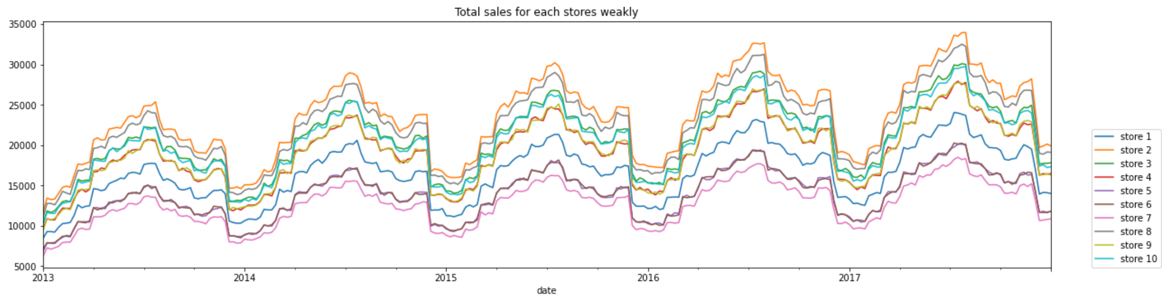


The histograms of sales of each stores shows how frequently a value falls into a particular bin. These are right skewed as many of the total sales values are near the lower end of the range, and higher sales values are infrequent.


Total sales for each stores yearly
1e6
1.4
1.2
1.0
0.8
0.6
2013
2014
2015
2016
2017
date
store 1
store 2
store 3
store 4
store 5
store 6
store 7
store 8
store 9
store 10


Total sales for each stores Quarterly
400000
350000
300000
250000
200000
150000
100000
2013
2014
2015
2016
2017
date
store 1
store 2
store 3
store 4
store 5
store 6
store 7
store 8
store 9
store 10


Total sales for each stores monthly
140000
120000
100000
80000
60000
40000
2013
2014
2015
2016
2017
date
store 1
store 2
store 3
store 4
store 5
store 6
store 7
store 8
store 9
store 10



From the year wise sales, there is an increasing trend from 2013 till the end for each stores where the highest sales occurred in store 2 and the least sales occurred in store 7. In the above plot of trends, it can be seen that there is a significant increase in the sales of stores in first quarter (between Jan and march). Then there is a small decrease after second quarter till fourth. In the end, we can see a decreasing trend. In the above plot of total sales, it can be seen that for every stores there is an increasing trend in the periods of January to July. Then we can see a decreasing trend till December with a slight exception on November. Even though the total sales of each stores varies from week to week, it shows us some interesting seasonal trends. In the above plots, we can see that there is a clear seasonality happening in the sales data for each year.

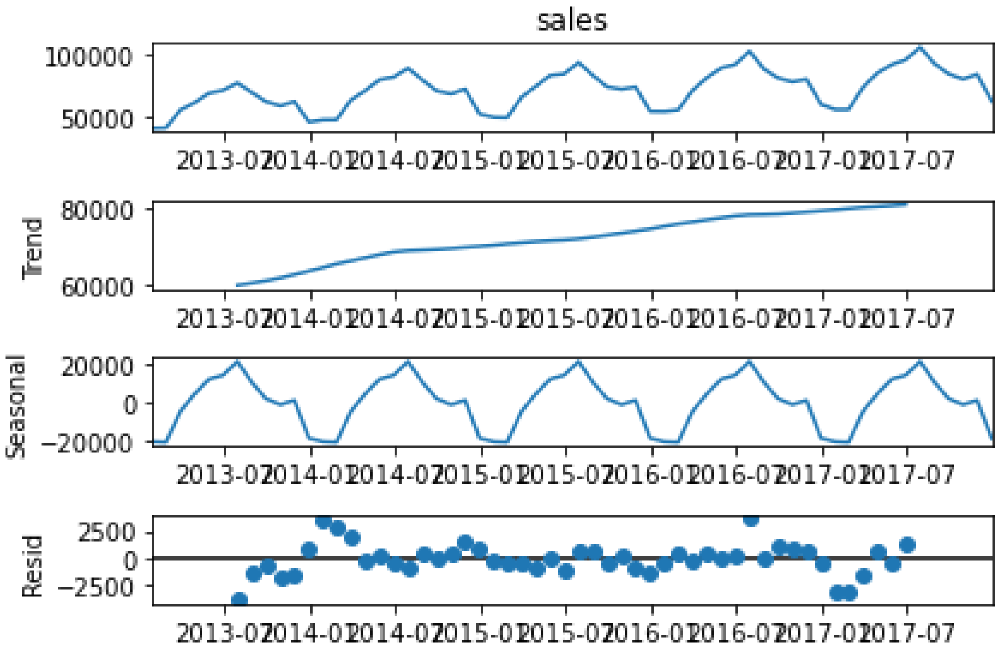
Time series decomposition — ETS model

ETS stands for **Error-Trend-Seasonality** and is a model used for the time series decomposition which is the breaking down of the series into its trend, seasonality and noise components. It helps in understanding the time series data better while using it to analyze and forecast. It is a uni-variate forecasting model used when dealing with time-series data. It focuses on trend and seasonal components. It has a seasonal_decompose function which takes the column name for which the season has to be decomposed and the type of seasonal component. It can be additive or multiplicative. By seeing the graph we can see that it is linear therefore the type has to be additive. These components are defined as follows:

Level: The average value in the series.
Trend: The increasing or decreasing value in the series.
Seasonality: The repeating short-term cycle in the series.
Noise: The random variation in the series.

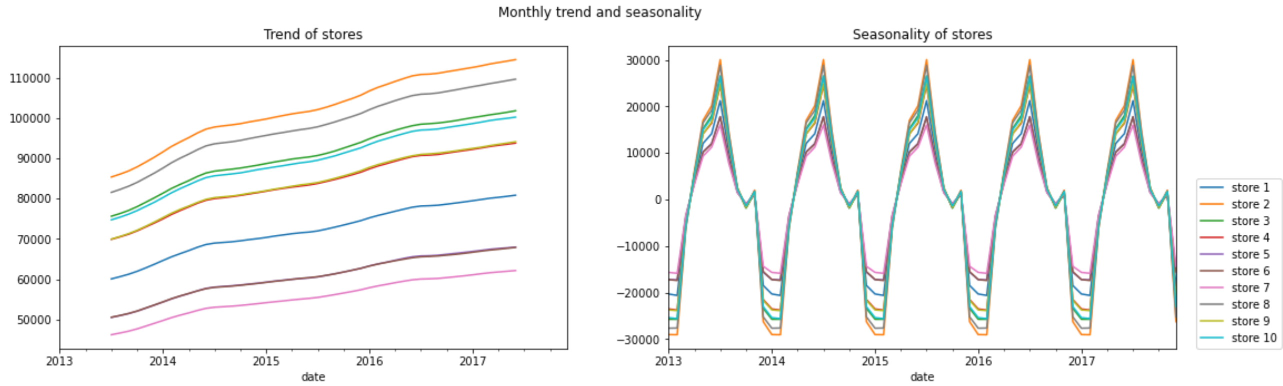
Running the ETS decomposition creates the series, performs the decomposition, and plots the 4 resulting series as shown in the below example of store 1's decomposition.

Here we can see that the entire series was taken as the trend component and also there was some seasonality.

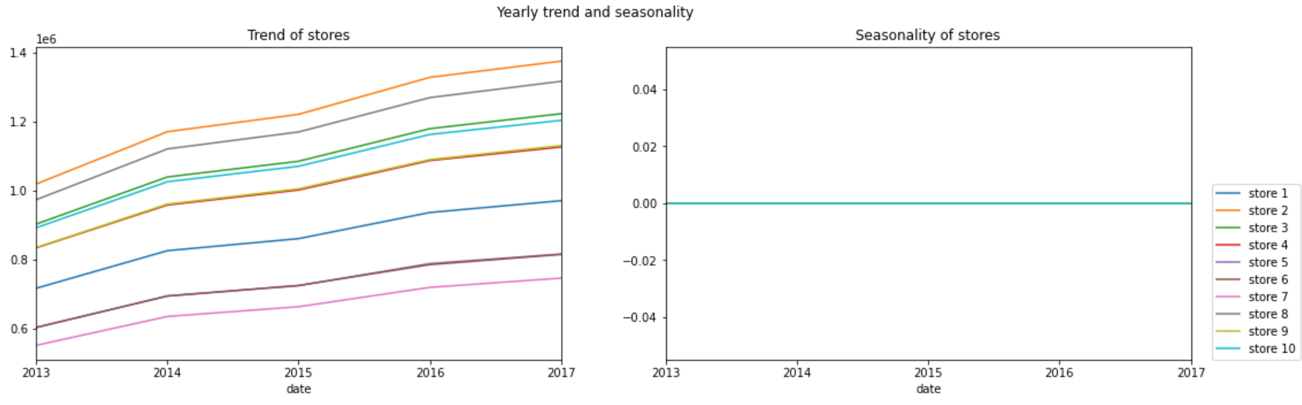


Here we can see that the range of trend and residual is nominal, or we can say that trend is having variation between 6000 to 8000, and most of the time residual is having the variation around. But for the seasonality, we can see that it varies between -2000 to 2000, which is a high difference range.

Running the example plots the observed, trend, seasonal, and residual time series. We can see that the trend and seasonality information extracted from the series does seem reasonable. The residuals are also interesting, showing periods of high variability in the early and later years of the series. In particular we can plot the monthly trends and seasonality of each stores together to identify the common behaviour of ten stores from below.



But if we look at yearly,we can see that there is no seasonality for any of the stores and there is a trend component.



Test for stationarity

Stationarity implies the statistical properties of a time series do not change over time. A time series is said to be “stationary” if it has no trend, exhibits constant variance over time, and has a constant autocorrelation structure over time. Most often, it happens when the data is non-stationary the predictions we get from the ARIMA model are worse or not that accurate. If the data is not stationary, we can do one thing: either make the data stationary or use the SARIMAX model.

To know more about the time series stationarity, we can perform the ADfuller test or KPSS Test, tests based on hypothesis.

1. Augmented Dickey-Fuller Test:

H0: The time series is non-stationary. In other words, it has some time-dependent structure and does not have constant variance over time.

HA: The time series is stationary.

If the p-value from the test is less than some significance level (e.g. α = .05), then we can reject the null hypothesis and conclude that the time series is stationary, where if the p-value is less than 0.05, then we can consider the time series is

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stationary, and if the P-value is greater than 0.05, then the time series is non-stationary.

Performing the adfuller test on data:-

Stationarity of store_1

Augmented Dickey-Fuller Test:
ADF test statistic -4.995500
p-value 0.000023
lags used 11.000000
observations 48.000000
critical value (1%) -3.574589
ADF test statistic -4.995500
p-value 0.000023
lags used 11.000000
observations 48.000000
critical value (1%) -3.574589
critical value (5%) -2.923954
ADF test statistic -4.995500
p-value 0.000023
lags used 11.000000
observations 48.000000
critical value (1%) -3.574589
critical value (5%) -2.923954
critical value (10%) -2.600039
Strong evidence against the null hypothesis
Reject the null hypothesis
Data has no unit root and is stationary

Stationarity of store_2

Augmented Dickey-Fuller Test:
ADF test statistic -4.895429
p-value 0.000036
lags used 11.000000
observations 48.000000
critical value (1%) -3.574589
ADF test statistic -4.895429
p-value 0.000036
lags used 11.000000
observations 48.000000
critical value (1%) -3.574589
critical value (5%) -2.923954
ADF test statistic -4.895429
p-value 0.000036
lags used 11.000000
observations 48.000000
critical value (1%) -3.574589
critical value (5%) -2.923954
critical value (10%) -2.600039
Strong evidence against the null hypothesis
Reject the null hypothesis

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Data has no unit root and is stationary
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Stationarity of store_3

Augmented Dickey-Fuller Test:
ADF test statistic     -5.119738
p-value                0.000013
# lags used            11.000000
# observations         48.000000
critical value (1%)    -3.574589
ADF test statistic     -5.119738
p-value                0.000013
# lags used            11.000000
# observations         48.000000
critical value (1%)    -3.574589
critical value (5%)    -2.923954
ADF test statistic      -5.119738
p-value                 0.000013
# lags used             11.000000
# observations          48.000000
critical value (1%)     -3.574589
critical value (5%)     -2.923954
critical value (10%)    -2.600039
Strong evidence against the null hypothesis
Reject the null hypothesis
Data has no unit root and is stationary
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Stationarity of store_4

Augmented Dickey-Fuller Test:
ADF test statistic     -5.321706
p-value                0.000005
# lags used            11.000000
# observations         48.000000
critical value (1%)    -3.574589
ADF test statistic     -5.321706
p-value                0.000005
# lags used            11.000000
# observations         48.000000
critical value (1%)    -3.574589
critical value (5%)    -2.923954
ADF test statistic      -5.321706
p-value                 0.000005
# lags used             11.000000
# observations          48.000000
critical value (1%)     -3.574589
critical value (5%)     -2.923954
critical value (10%)    -2.600039
Strong evidence against the null hypothesis
Reject the null hypothesis
Data has no unit root and is stationary
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Stationarity of store_5

Augmented Dickey-Fuller Test:
ADF test statistic     -4.982077
p-value                0.000024
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lags used 11.000000
observations 48.000000
critical value (1%) -3.574589
ADF test statistic -4.982077
p-value 0.000024
lags used 11.000000
observations 48.000000
critical value (1%) -3.574589
critical value (5%) -2.923954
ADF test statistic -4.982077
p-value 0.000024
lags used 11.000000
observations 48.000000
critical value (1%) -3.574589
critical value (5%) -2.923954
critical value (10%) -2.600039
Strong evidence against the null hypothesis
Reject the null hypothesis
Data has no unit root and is stationary

Stationarity of store_6

Augmented Dickey-Fuller Test:
ADF test statistic -5.435734
p-value 0.000003
lags used 11.000000
observations 48.000000
critical value (1%) -3.574589
ADF test statistic -5.435734
p-value 0.000003
lags used 11.000000
observations 48.000000
critical value (1%) -3.574589
critical value (5%) -2.923954
ADF test statistic -5.435734
p-value 0.000003
lags used 11.000000
observations 48.000000
critical value (1%) -3.574589
critical value (5%) -2.923954
critical value (10%) -2.600039
Strong evidence against the null hypothesis
Reject the null hypothesis
Data has no unit root and is stationary

Stationarity of store_7

Augmented Dickey-Fuller Test:
ADF test statistic -5.440426
p-value 0.000003
lags used 11.000000
observations 48.000000
critical value (1%) -3.574589
ADF test statistic -5.440426
p-value 0.000003
lags used 11.000000
observations 48.000000

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critical value (1%)   -3.574589
critical value (5%)   -2.923954
ADF test statistic   -5.440426
p-value   0.000003
lags used   11.000000
observations   48.000000
critical value (1%)   -3.574589
critical value (5%)   -2.923954
critical value (10%)   -2.600039
Strong evidence against the null hypothesis
Reject the null hypothesis
Data has no unit root and is stationary

Stationarity of store_8

Augmented Dickey-Fuller Test:
ADF test statistic   -5.235074
p-value   0.000007
lags used   11.000000
observations   48.000000
critical value (1%)   -3.574589
ADF test statistic   -5.235074
p-value   0.000007
lags used   11.000000
observations   48.000000
critical value (1%)   -3.574589
critical value (5%)   -2.923954
ADF test statistic   -5.235074
p-value   0.000007
lags used   11.000000
observations   48.000000
critical value (1%)   -3.574589
critical value (5%)   -2.923954
critical value (10%)   -2.600039
Strong evidence against the null hypothesis
Reject the null hypothesis
Data has no unit root and is stationary

Stationarity of store_9

Augmented Dickey-Fuller Test:
ADF test statistic   -5.385964
p-value   0.000004
lags used   11.000000
observations   48.000000
critical value (1%)   -3.574589
ADF test statistic   -5.385964
p-value   0.000004
lags used   11.000000
observations   48.000000
critical value (1%)   -3.574589
critical value (5%)   -2.923954
ADF test statistic   -5.385964
p-value   0.000004
lags used   11.000000
observations   48.000000
critical value (1%)   -3.574589

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critical value (5%)   -2.923954
critical value (10%)   -2.600039
Strong evidence against the null hypothesis
Reject the null hypothesis
Data has no unit root and is stationary

Stationarity of store_10

Augmented Dickey-Fuller Test:
ADF test statistic   -5.144744
p-value   0.000011
lags used   11.000000
observations   48.000000
critical value (1%)   -3.574589
ADF test statistic   -5.144744
p-value   0.000011
lags used   11.000000
observations   48.000000
critical value (1%)   -3.574589
critical value (5%)   -2.923954
ADF test statistic   -5.144744
p-value   0.000011
lags used   11.000000
observations   48.000000
critical value (1%)   -3.574589
critical value (5%)   -2.923954
critical value (10%)   -2.600039
Strong evidence against the null hypothesis
Reject the null hypothesis
Data has no unit root and is stationary

Here we can see that the p-value is less for every stores, and we can say that the evidence of the null hypothesis is high; hence the time series is stationary. (Here we take monthly variation of the data. But when considering yearly every stores are non stationary and for quarterly every stores except store 3 are non stationary.)

2. Kwiatkowski-Phillips-Schmidt-Shin ("KPSS") test:-

Null Hypothesis: The process is trend stationary.

Alternate Hypothesis: The series has a unit root (series is not stationary).

If the test statistic is greater than the critical value, we reject the null hypothesis (series is not stationary).

Performing KPSS Test:

Stationarity of store_1

Results of KPSS Test:
Test Statistic 0.538538
p-value 0.032987
Lags Used 4.000000
Critical Value (10%) 0.347000
Critical Value (5%) 0.463000
Critical Value (2.5%) 0.574000
Critical Value (1%) 0.739000
dtype: float64
Weak evidence against the null hypothesis
Fail to reject the null hypothesis
Data has a unit root and is stationary

Stationarity of store_2

Results of KPSS Test:
Test Statistic 0.534764
p-value 0.033837
Lags Used 4.000000
Critical Value (10%) 0.347000
Critical Value (5%) 0.463000
Critical Value (2.5%) 0.574000
Critical Value (1%) 0.739000
dtype: float64
Weak evidence against the null hypothesis
Fail to reject the null hypothesis
Data has a unit root and is stationary

Stationarity of store_3

Results of KPSS Test:
Test Statistic 0.539060
p-value 0.032869
Lags Used 4.000000
Critical Value (10%) 0.347000
Critical Value (5%) 0.463000
Critical Value (2.5%) 0.574000
Critical Value (1%) 0.739000
dtype: float64
Weak evidence against the null hypothesis
Fail to reject the null hypothesis
Data has a unit root and is stationary

Stationarity of store_4

Results of KPSS Test:
Test Statistic 0.535058
p-value 0.033771
Lags Used 4.000000
Critical Value (10%) 0.347000
Critical Value (5%) 0.463000
Critical Value (2.5%) 0.574000
Critical Value (1%) 0.739000
dtype: float64
Weak evidence against the null hypothesis
Fail to reject the null hypothesis
Data has a unit root and is stationary

Stationarity of store_5

Results of KPSS Test:
Test Statistic 0.537641
p-value 0.033189
Lags Used 4.000000
Critical Value (10%) 0.347000
Critical Value (5%) 0.463000
Critical Value (2.5%) 0.574000
Critical Value (1%) 0.739000
dtype: float64
Weak evidence against the null hypothesis
Fail to reject the null hypothesis
Data has a unit root and is stationary

Stationarity of store_6

Results of KPSS Test:
Test Statistic 0.531042
p-value 0.034675
Lags Used 4.000000
Critical Value (10%) 0.347000
Critical Value (5%) 0.463000
Critical Value (2.5%) 0.574000
Critical Value (1%) 0.739000
dtype: float64
Weak evidence against the null hypothesis
Fail to reject the null hypothesis
Data has a unit root and is stationary

Stationarity of store_7

Results of KPSS Test:
Test Statistic 0.537649
p-value 0.033187
Lags Used 4.000000
Critical Value (10%) 0.347000
Critical Value (5%) 0.463000
Critical Value (2.5%) 0.574000
Critical Value (1%) 0.739000
dtype: float64
Weak evidence against the null hypothesis
Fail to reject the null hypothesis
Data has a unit root and is stationary

Stationarity of store_8

Results of KPSS Test:
Test Statistic 0.536610
p-value 0.033421
Lags Used 4.000000
Critical Value (10%) 0.347000
Critical Value (5%) 0.463000
Critical Value (2.5%) 0.574000
Critical Value (1%) 0.739000
dtype: float64

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Weak evidence against the null hypothesis
Fail to reject the null hypothesis
Data has a unit root and is stationary
----------------------------------------------------------------------------------------------
Stationarity of store_9

Results of KPSS Test:
Test Statistic           0.538034
p-value                  0.033100
Lags Used                4.000000
Critical Value (10%)     0.347000
Critical Value (5%)      0.463000
Critical Value (2.5%)    0.574000
Critical Value (1%)      0.739000
dtype: float64
Weak evidence against the null hypothesis
Fail to reject the null hypothesis
Data has a unit root and is stationary
----------------------------------------------------------------------------------------------
Stationarity of store_10

Results of KPSS Test:
Test Statistic           0.531977
p-value                  0.034465
Lags Used                4.000000
Critical Value (10%)     0.347000
Critical Value (5%)      0.463000
Critical Value (2.5%)    0.574000
Critical Value (1%)      0.739000
dtype: float64
Weak evidence against the null hypothesis
Fail to reject the null hypothesis
Data has a unit root and is stationary
----------------------------------------------------------------------------------------------
```

Hence from both of the above results we can summarize as our data is stationary for every stores when dealing it monthly.

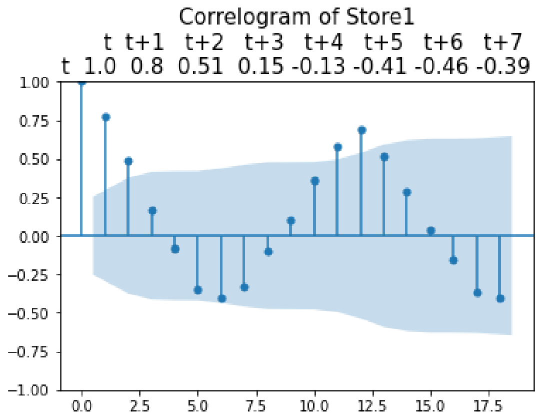
Autocorrelation

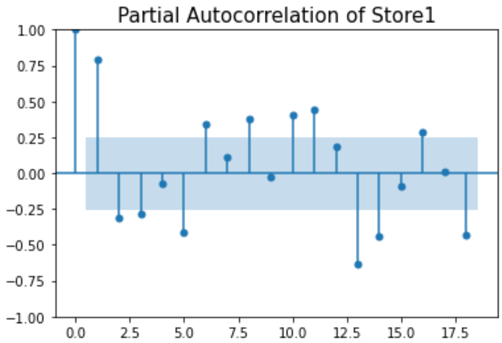
Autocorrelation function (ACF) and Partial Autocorrelation Function (PACF, also called Partial ACF) are important functions in analyzing a time series. They generally produce plots that are very important in finding the values p, q and r for Autoregressive (AR) and Moving Average (MA) models. The autocorrelation analysis helps detect patterns and check for randomness.

An ACF measures and plots the average correlation between data points in time series and previous values of the series measured for different lag lengths. A PACF is similar to an ACF except that each partial correlation controls for any correlation between observations of a shorter lag length.

A plot of the autocorrelation of a time series by lag is called the AutoCorrelation Function (ACF) or correlogram. A correlogram plots the correlation of all possible timesteps. The lagged variables with the highest correlation can be considered for modeling.

Here we have our ACF and PACF of our data.





Both the ACF and PACF start with a lag of 0, which is the correlation of the time series with itself and therefore results in a correlation of 1. Additionally, there is a blue area in the ACF and PACF plots. This blue area depicts the 95% confidence interval and is an indicator of the significance threshold. That means, anything within the blue area is statistically close to zero and anything outside the blue area is statistically non-zero. There are several autocorrelations that are significantly non-zero. Therefore, the time series is non-random. Also there is a high degree of autocorrelation between adjacent (lag = 1) in PACF plot.

Similarly we have ACF and PACF plots of other stores as well.

Time series forecasting with SARIMAX

We use the statsmodels SARIMAX package to train the model and generate dynamic predictions. Here it is the SARIMAX results summary when the model is fitted to the order (0,1,0) and seasonal order (0,1,0,12).

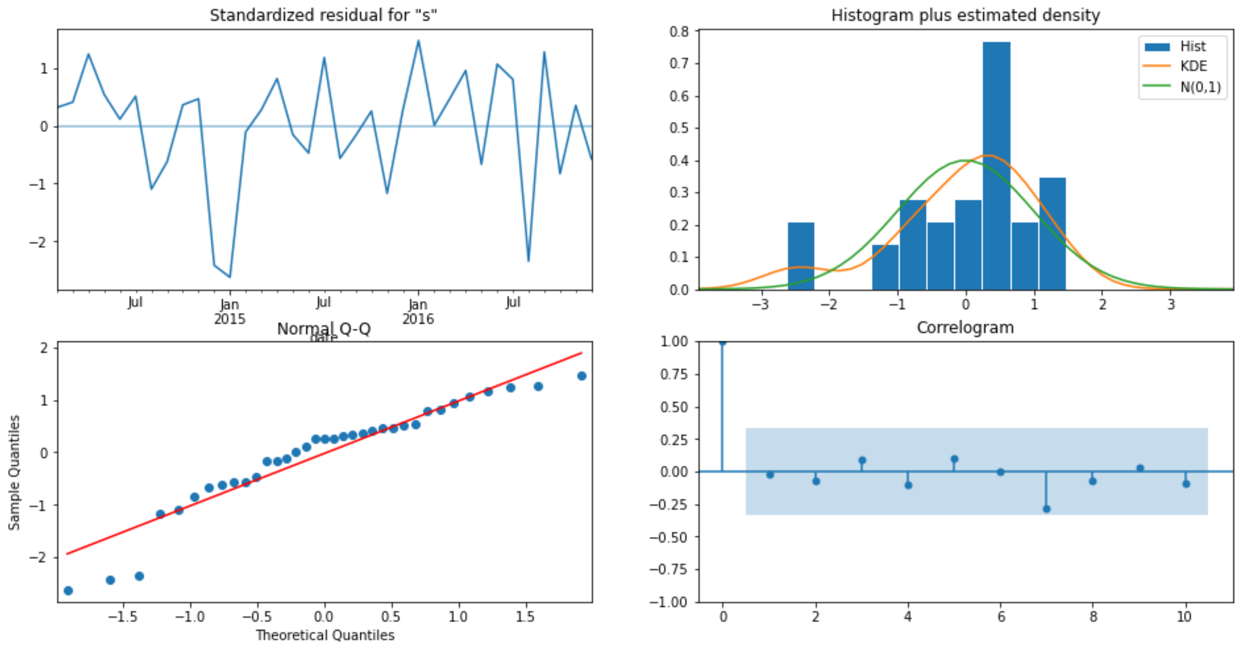
SARIMAX Results

Dep. Variable:	sales	No. Observations:	48
Model:	SARIMAX(0, 1, 0)x(0, 1, 0, 12)	Log Likelihood	-188.830
Date:	Fri, 17 Feb 2023	AIC	379.660
Time:	22:57:00	BIC	381.215
Sample:	01-31-2013	HQIC	380.197
	- 12-31-2016		
Covariance Type:	opg		

	coef	std err	z	P>|z|	[0.025	0.975]
sigma2	2842.2227	581.269	4.890	0.000	1702.957	3981.489

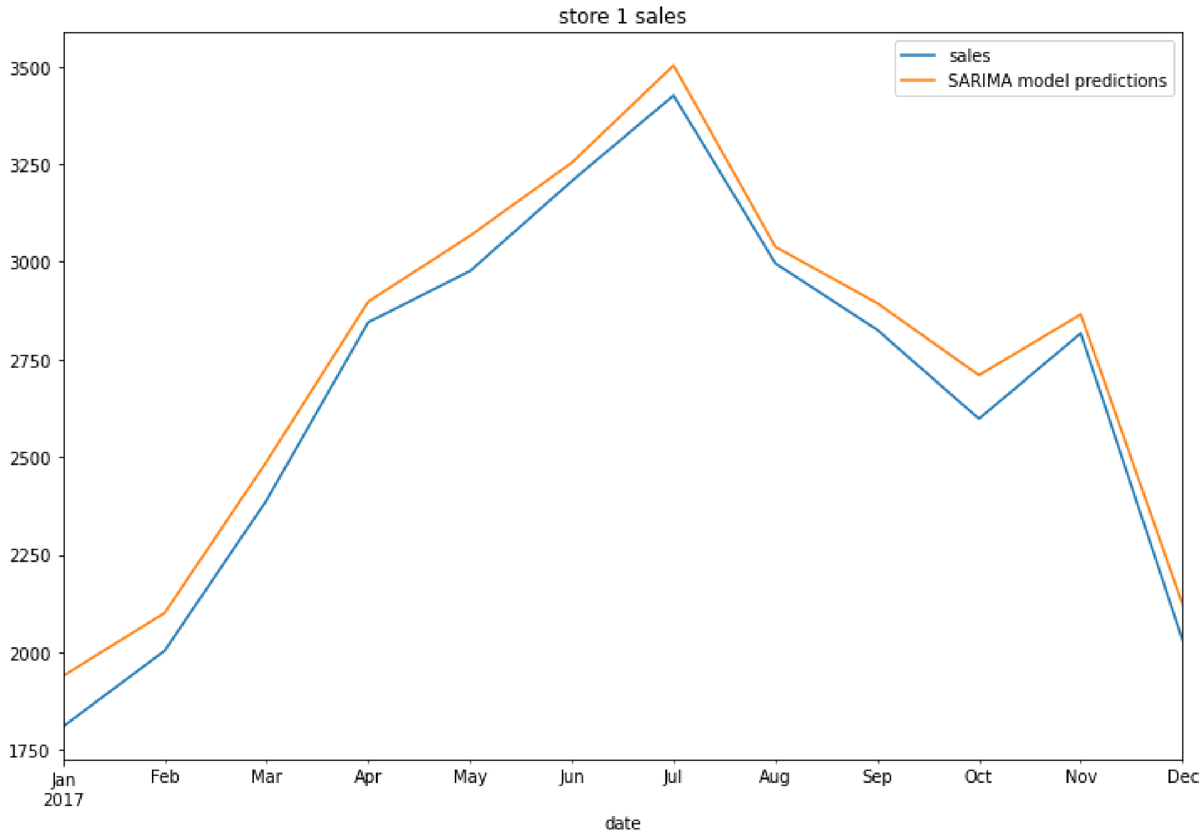
Ljung-Box (L1) (Q):	0.02	Jarque-Bera (JB):	6.37
Prob(Q):	0.90	Prob(JB):	0.04
Heteroskedasticity (H):	0.81	Skew:	-0.99
Prob(H) (two-sided):	0.72	Kurtosis:	3.65

Plot obtained after running model diagnostics test on the SARIMA model built for store 1 data:



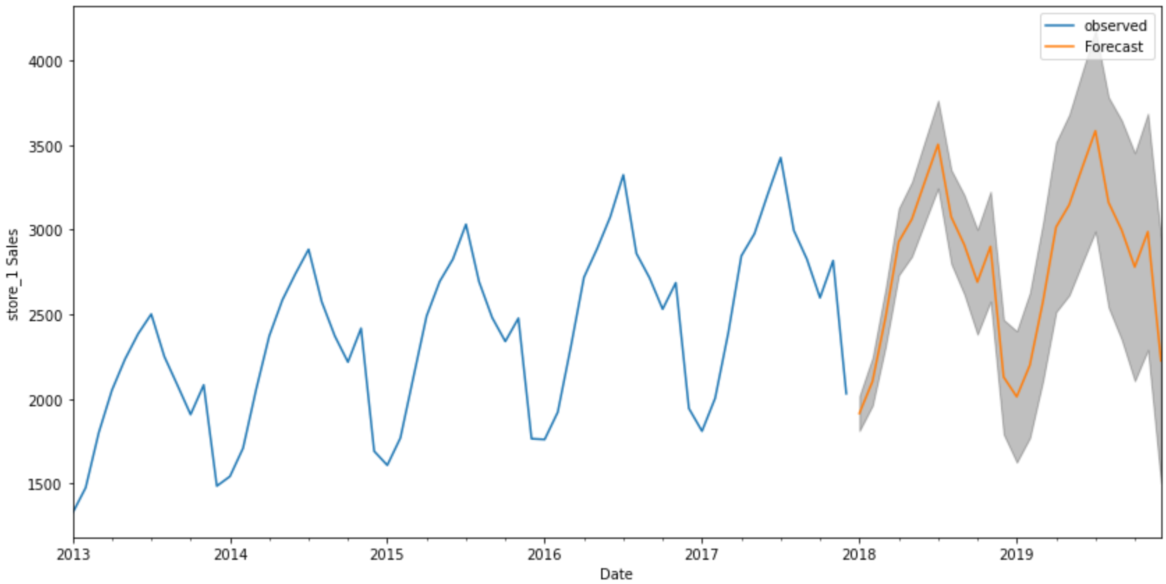
From the above plots, we can say that our residuals are normally distributed.

Plot obtained after making predictions about the known future and comparing it to the data points in the test set of store 1 sales:



From the above plot, we can say that we have a good model, which had predicted better in the majority of months. The predicted results have a same kind of trend of the real sales data as well. Here in the graph, we can see that the forecasting line is almost lying on the given values for this model. We didn't even require the differencing method. Using this model now, we can predict the future values too.

Plot obtained after making predictions to the unknown future for store 1 sales (predicted values for the next two years):



In the above plot, the forecasted values for the next two years (2018 and 2019) have been plotted. The plot tells us that the higher number of sales would be occurring in June/July for the years 2018 and 2019. In the forecasting we can see that the sales are increasing better first which tells us that the demand for office supplies will increase during these months and then it has gradual decrease after six months. We can see that the model has predicted the values without compromising with the seasonality effects and exogenous factors. And the trend line is almost going as usual as it was going in past years.

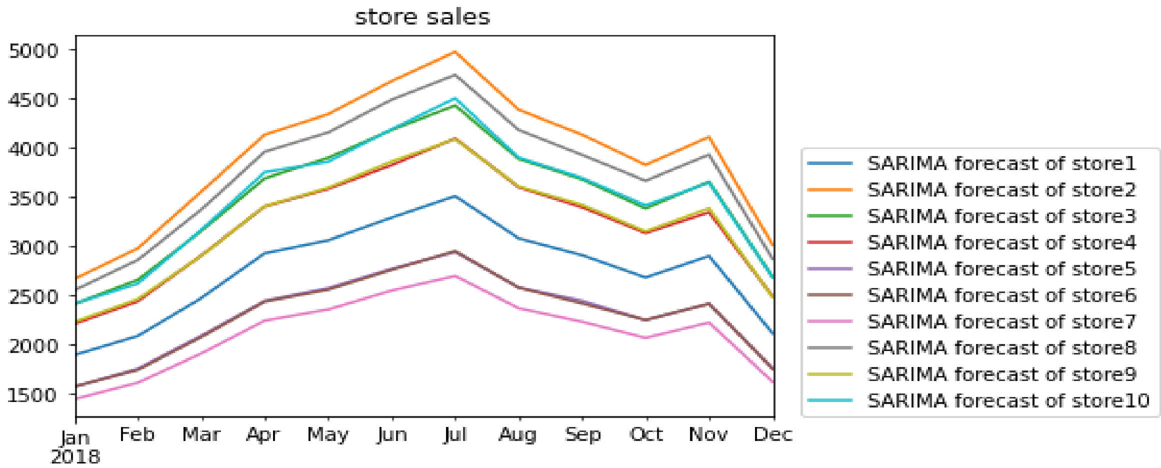
Comparing the scores of SARIMAX models in sales of ten stores

Similarly we can train the model and generate dynamic predictions for other stores too. Here it is the root mean square error values of ten stores for SARIMAX model fitting.

---

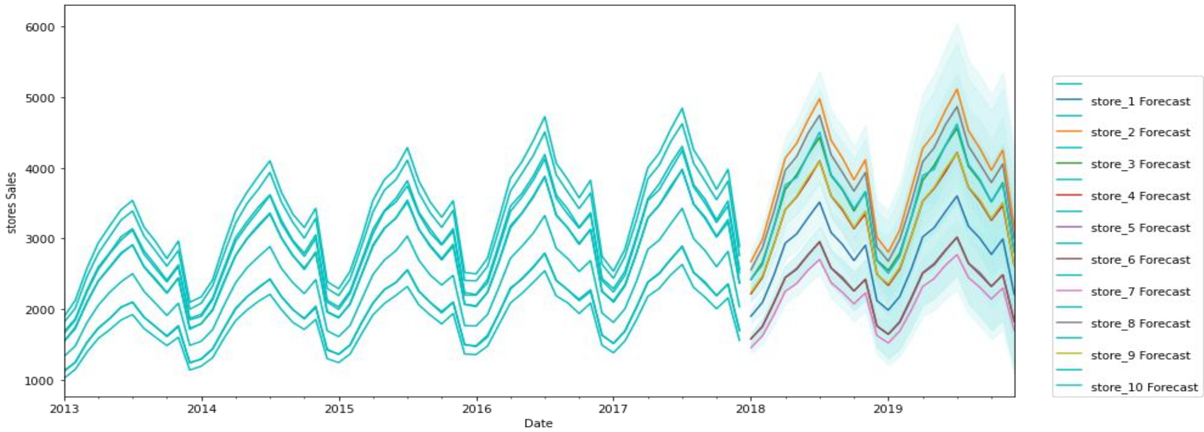
rmse of store1: 84.00605349899561
rmse of store2: 99.89134081126008
rmse of store3: 79.34385672449186
rmse of store4: 77.54930681293042
rmse of store5: 61.16243642312346
rmse of store6: 52.507506234600314
rmse of store7: 52.03590672667766
rmse of store8: 100.49471951371982
rmse of store9: 63.818170080352594
rmse of store10: 40.117617326087625

Plot comparing the data points in monthly sales of stores



From the above plot it is clear that for majority of the months, sales of the second store is higher than other stores. But overall trend of each stores are same. There is an increasing trend in the periods of January to July. Then we can see a decreasing trend till December with a slight exception on November.

Plot made after forecasting sales values of stores for next two years:

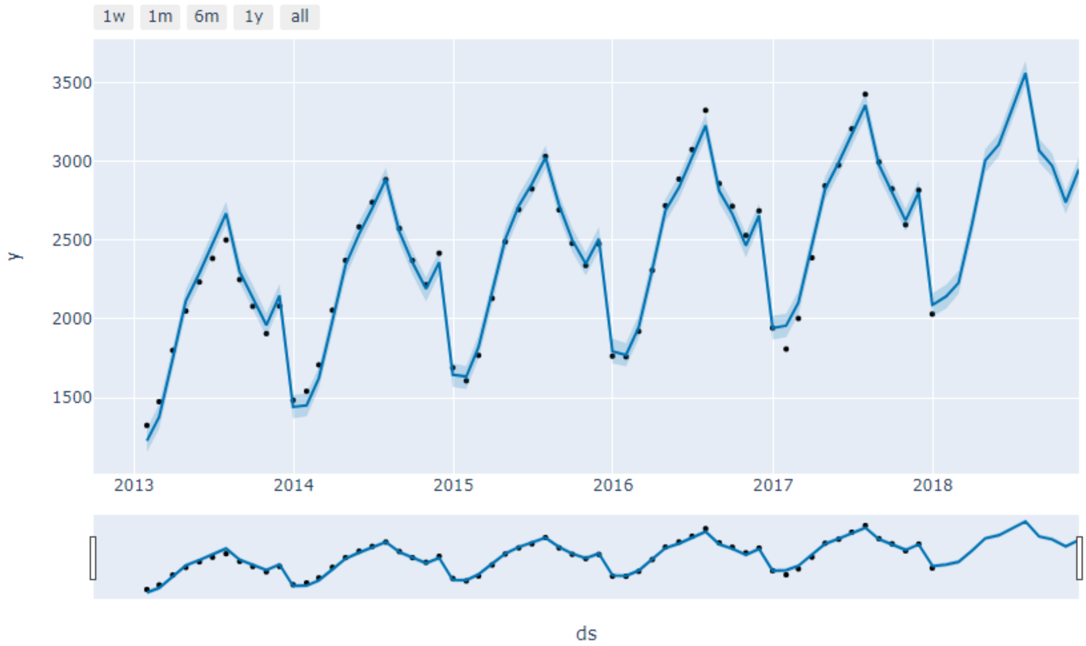


Plots made with the help of fbprophet library

Plot made after forecasting sales values of store 1 for next two years:

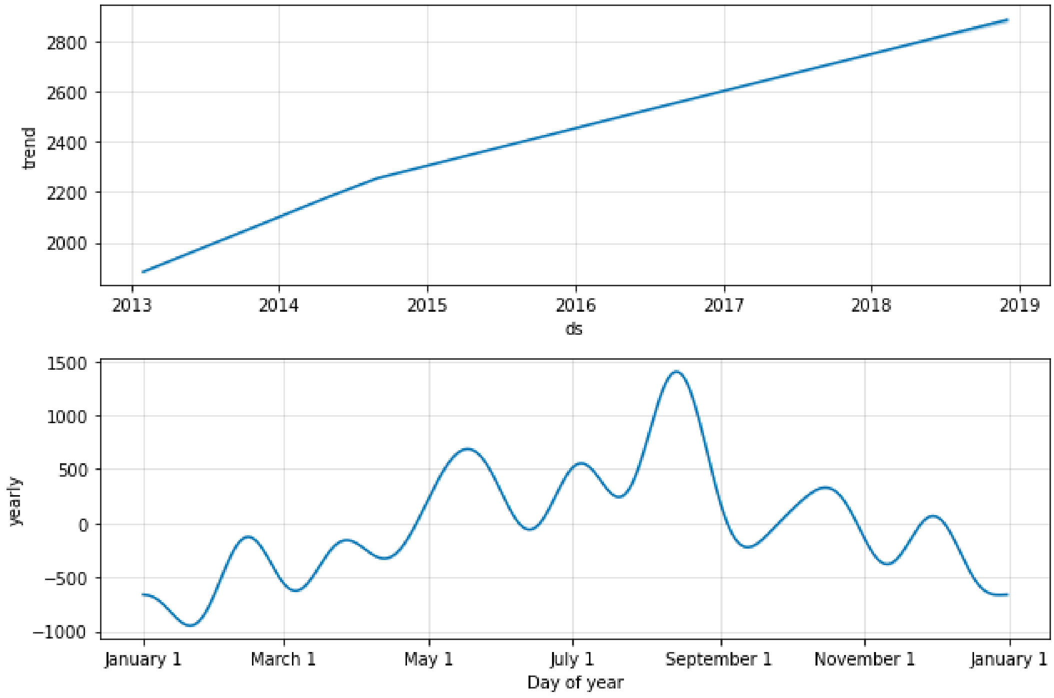
In the predicted values, highest sales are predicted during the month of June/July 2019 and the lowest sales are predicted during the month of January 2013. Generally sales has an increasing year wise trend.

We can see that in every years, the model has predicted more sales during the months of June and July while comparing to other months. The model has predicted lowest sales in the month of January and December in every years.



From the plot of trends, it is clear that the trend is linearly increasing over the years. So, the sales of the store 1 products are increasing over the years that means demand of the store 1 products are increasing over the years.

From the plot of yearly seasonality, it is clear that higher seasonality occurs during the middle part of August and then in the starting of June while the lowest seasonality occurs during January and February.



Comparing the scores of Prophet models in sales of ten stores

Similarly we can find the root mean square error values of ten stores for Prophet model fitting.

rmse of store1: 57.33522247727067
rmse of store2: 81.51222610487261
rmse of store3: 70.65213367116996
rmse of store4: 64.6961694933252
rmse of store5: 48.248088606450786
rmse of store6: 48.16648379380419
rmse of store7: 43.65374044889431
rmse of store8: 75.96172058834054
rmse of store9: 64.79600460681706
rmse of store10: 69.01518166887772

Challenges & Opportunities

- I was able to create SARIMA models for each stores sales data.
- I also tried to build prophet models for this data.
- I was able to plot useful visualizations of the time series forecasting.

Reflections on the Internship

It was a great learning experience. It gave me an idea on how projects are done in the industry. Submitting daily activity reports helped me to keep track of what I was doing each day.

Conclusions

- From the store wise sales, the highest sales occurred in store 2 and the least sales occurred in store 7.
- There are some items with a high sale as item 15 or 29 and also there are some with very low sale as item 3 or 41. So the manager should adopt any new business approaches to increase the sales of the items.
- There is an increasing trend from 2013 till the end for each stores. So every year the business got higher.
- There is a significant increase in the sales of stores in first quarter (between Jan and march). Then there is a small decrease after second quarter till fourth. In the end, we can see a decreasing trend. So the sale's team should concentrate on the second half of every years.
- For every stores there is an increasing trend in the periods of January to July. Then we can see a decreasing trend till December with a slight exception on November. Even though the total sales of each stores varies from week to week, it shows us some interesting seasonal trends.
- The given time series is stationary.
- In every store, we can expect that the sale will increase in the next two years.
- In store 1, the sales may increase up to 3500 (expected during June 2018) and may decrease up to 2200 (Expected during December 2018).
- From all the models we can say that this sales generally has low sales during the month of January. More sales usually happen in the months of June and July.
- For the coming two years (2018 and 2019), more business will happen in store 2 than the other 9 stores and less sales in store 7.

References

- Introduction to Time Series Analysis and Forecasting - Douglas C. Montgomery ,Cheryl L. Jennings ,Murat Kulahci
- Introduction to Time Series and Forecasting - Peter J. Brockwell , Richard A. Davis
- http://home.ubalt.edu/ntsbarsh/Business-stat/stat-data/Forecast.html
- https://ai.plainenglish.io/time-series-decomposition-ets-model-using-python-4d2cd04bab77#:~:text=ETS%20stands%20for%20Error%2DTrend,on%20trend%20and%20seasonal%20components.
- https://analyticsindiamag.com/complete-guide-to-sarimax-in-python-for-time-series-modeling/
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Link to code and executable file: